

AMBLER, AK, USA

WMO: 701718

Lat: 67.100N

Lon: 157.850W

Elev: 88

StdP: 100.27

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26551

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.8	-37.6	-34.2	0.2	-28.4	-33.8	0.2	-27.6	9.6	-15.8	9.0	-15.3	0.8	40	0.738

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	24.9	15.3	22.9	14.4	21.3	13.8	16.5	22.9	15.4	21.2	14.5	19.8	2.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	9.9	18.0	12.6	9.2	17.2	12.1	8.9	16.7	46.3	22.8	43.1	21.5	40.5	19.7	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.6	6.8	DB	-42.5	26.6	4.8	2.5	-46.0	28.4	-48.7	29.8	-51.4	31.2	-54.9	33.0	
			WB	-36.0	17.6	4.0	1.3	-38.9	18.5	-41.3	19.3	-43.5	20.1	-46.5	21.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.2	-19.1	-15.1	-14.4	-5.5	4.5	13.6	14.7	11.2	5.9	-4.7	-13.5	-16.8
	DBStd	13.93	10.66	8.73	7.43	6.78	5.94	3.90	3.56	3.56	4.11	6.51	7.73	8.32
	HDD10.0	5180	903	703	755	466	185	12	6	27	131	457	706	829
	HDD18.3	7878	1162	937	1013	716	428	147	122	222	372	715	956	1088
	CDD10.0	357	0	0	0	0	16	121	150	63	8	0	0	0
	CDD18.3	14	0	0	0	0	0	6	8	1	0	0	0	0
	CDH23.3	139	0	0	0	0	2	59	68	9	0	0	0	0
	CDH26.7	17	0	0	0	0	0	8	9	0	0	0	0	0
Wind	WSAvg	2.8	2.9	3.3	3.1	3.3	2.8	2.3	2.0	2.1	2.7	2.8	2.9	3.1
Precipitation	PrecAvg	457	32	33	24	19	20	33	59	70	49	45	38	34
	PrecMax	575	69	92	79	57	42	66	106	157	93	115	85	76
	PrecMin	338	7	6	4	2	7	13	30	26	25	19	5	15
	PrecStd	65	16	20	15	13	9	13	18	27	20	21	23	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.1	2.5	10.5	22.7	27.7	27.8	24.8	18.5	8.9	2.6	1.3
		MCWB	-0.4	0.1	0.2	4.8	12.0	16.1	16.7	16.4	10.7	4.1	-0.1	-0.3
	2%	DB	-1.2	-0.1	0.3	7.5	19.1	24.9	25.2	21.9	15.5	6.5	1.2	-1.1
		MCWB	-1.9	-0.9	-1.1	2.8	10.7	14.6	15.9	15.1	9.5	3.6	-0.6	-1.9
	5%	DB	-2.7	-2.1	-2.1	5.8	16.3	22.8	23.0	18.9	13.0	4.6	-1.1	-2.9
		MCWB	-3.8	-3.1	-3.7	1.7	9.0	13.6	15.1	13.4	9.0	2.5	-1.9	-4.2
	10%	DB	-4.5	-3.7	-3.7	3.2	13.3	21.3	22.0	17.3	11.9	2.9	-2.7	-6.0
		MCWB	-5.4	-4.9	-5.4	0.4	7.6	13.0	14.8	12.3	8.6	1.1	-4.0	-6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.3	0.8	4.8	12.7	17.0	17.9	17.6	11.9	5.9	1.1	0.1
		MCDB	0.7	1.0	1.9	9.7	21.7	25.6	25.2	23.4	15.2	7.5	2.1	0.8
	2%	WB	-1.8	-0.9	-1.1	3.1	11.1	15.4	16.8	15.6	10.7	4.2	-0.3	-1.9
		MCDB	-1.3	-0.2	0.2	6.5	18.4	22.9	23.5	19.9	13.7	6.3	0.6	-1.4
	5%	WB	-3.6	-2.7	-3.3	2.1	9.7	14.5	16.0	14.2	9.9	2.8	-2.0	-4.0
		MCDB	-2.5	-2.0	-2.0	5.2	15.6	21.3	22.1	18.1	12.6	4.4	-1.5	-2.9
	10%	WB	-5.0	-4.4	-5.1	0.8	8.0	13.6	15.0	13.1	8.9	1.4	-4.0	-6.4
		MCDB	-3.9	-3.4	-3.5	3.3	12.9	20.1	20.4	16.4	11.4	3.0	-2.6	-5.4
Mean Daily Temperature Range	5% DB	MDBR	6.7	6.6	7.9	8.2	9.0	10.1	8.5	7.7	6.7	5.1	5.6	5.9
		MCDBR	5.8	5.6	7.2	8.8	13.3	12.5	11.5	10.7	9.2	5.8	4.3	5.7
		MCWBR	5.5	5.1	6.0	6.0	7.1	5.7	5.4	5.8	5.4	4.2	3.9	5.2
	5% WB	MCDBR	5.8	5.4	6.5	8.0	12.0	11.3	10.5	8.6	7.5	5.0	4.2	5.6
		MCWBR	5.6	5.0	5.5	5.5	6.6	5.5	5.2	5.1	4.5	3.9	4.1	5.2
Clear-Sky Solar Irradiance	taub	0.171	0.240	0.258	0.321	0.350	0.355	0.363	0.347	0.307	0.249	0.167	N/A	
	taud	1.913	2.116	2.240	2.129	2.266	2.386	2.387	2.448	2.508	2.309	1.937	N/A	
	Ebn at Noon	199	598	808	824	842	849	828	805	759	600	188	0	
	Edh at Noon	12	49	76	113	111	102	99	84	61	40	11	0	
All-Sky Solar Radiation	RadAvg	0.07	0.63	2.43	4.66	5.48	5.71	4.59	3.41	2.08	0.77	0.15	0.00	
	RadStd	0.01	0.06	0.17	0.24	0.56	0.51	0.39	0.43	0.26	0.11	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page